Higgott Chapter 7 source evidence

Oscar Higgott, Practical and Efficient Quantum Error Correction, UCL PhD. UCL record year 2024; title-page submission date 1 October 2023. Primary target: Figure 7.14a. [1]

Audit conclusion. The source supports comparing a 16-qubit Bolza code with four 24-qubit planar patches encoding the same four logical qubits: 96 / 16 = 6. This qubit-count ratio is arithmetic, not by itself a replication of the plotted error performance.

Critical unresolved choice. The figure caption says 16 rounds, implying 48 subrounds under the chapter's definition. Public circuit templates are labeled 96 subrounds. The accompanying report evaluates the duration choices; the source evidence appears on page 2.

Table 7.2 | Printed page 174 | PDF page 174 of 266 (1-based)

Table 7.2: Parameters of a family of semi-hyperbolic floquet codes derived from the Bolza surface. Here, n_{tot} is the total number of physical qubits (including ancillas) for the SD6 noise model. We verified that the EM3 distance d_e equals the embedded distance d reported here for all codes in the table.

l	k	n	d	kd^2/n	n_{tot}	d_s	kd_s^2/n_{tot}
1	4	16	2	1.00	40	2	0.40
2	4	64	3	0.56	160	4	0.40
3	4	144	4	0.44	360	6	0.40
4	4	256	6	0.56	640	8	0.40
5	4	400	7	0.49	1,000	10	0.40
6	4	576	8	0.44	1,440	12	0.40
7	4	784	10	0.51	1,960	14	0.40
8	4	1,024	11	0.47	2,560	16	0.40
9	4	1,296	12	0.44	3,240	18	0.40
10	4	1,600	14	0.49	4,000	20	0.40

What the table establishes

For refinement l = 1: k = 4, n = 16, d = 2. For l = 2: k = 4, n = 64, d = 3. The n_{tot} column includes SD6 ancillas; it is not the EM3 hardware-qubit count. The author reports equality between the embedded and EM3 distances for this table. [1, p. 174]

Distance qualification: the chapter reports circuit/noise-model-dependent graphlike distances from Stim.shortest_graphlike_error. A graphlike search result is not an unrestricted distance proof. [1, pp. 168-169]

Scope: this appendix documents source evidence and figure-derived numerical targets, rather than a replication result. Printed page numbers equal 1-based PDF page numbers throughout this appendix.

Noise model and duration

EM3 is a joint pair-Pauli / measurement-flip channel, not independent DEPOLARIZE2(p) plus a readout flip of probability p. Initialization and final readout have flip probability p / 2. [1, p. 164]

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7.2.5 Circuits and noise models

In this chapter we consider two different noise models, corresponding to the EM3 and SD6 noise models used in Ref. [95]. Both noise models are characterised by a noise strength parameter $p \in [0,1]$ (the physical error rate).

7.2.5.1 EM3 noise model

The EM3 noise model (entangling measurement 3-step cycle) assumes that two-qubit Pauli measurements are available natively in the platform (and hence no ancilla qubits are needed), as is the case for Majorana-based architectures [49, 156]. When implementing the measurement of a two-qubit Pauli operator $P^c \otimes P^c$, with probability p we insert an error chosen uniformly at random from the set $\{I,X,Y,Z\}^{\otimes 2} \times \{\text{flip}, \text{no flip}\}$. Here the two qubit Pauli error in $\{I,X,Y,Z\}^{\otimes 2}$ is applied immediately before the measurement, and the "flip" operation flips the outcome of the measurement. We assume each measurement takes a single time step, and hence each cycle of three sub-rounds takes three time steps. Initialisation of a qubit in the Z basis is followed by an X error, inserted with probability $p/2$. Similarly, the measurement of a data qubit in the Z basis is preceded by an X error, inserted with probability $p/2$. Single-qubit initialisation and measurement in the X or Y basis is achieved using noisy Z-basis initialisation or measurement and noiseless single-qubit Clifford gates. We do not need to define idling errors because data qubits are never idle in this noise model.

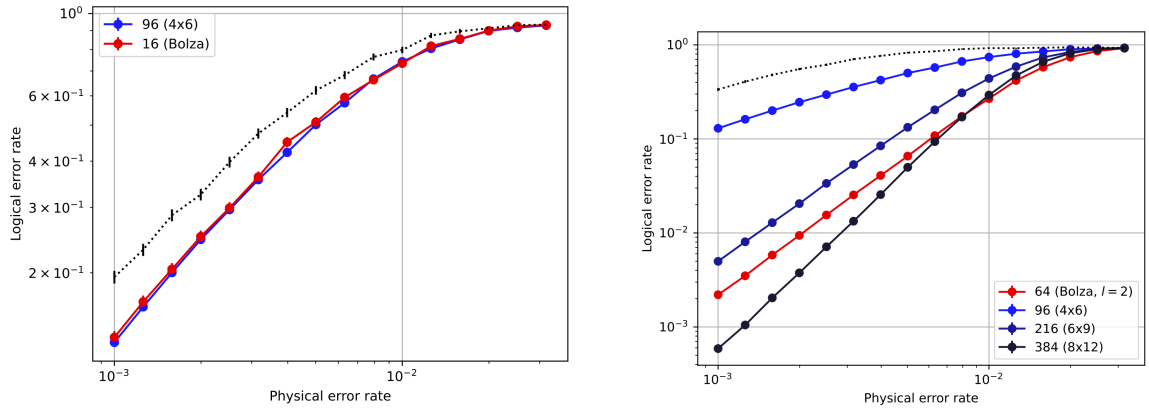
Figure 7.7 caption | Printed page 162 | PDF page 162 of 266 (1-based)

Figure 7.7: Each logical operator changes over a periodic sequence of six sub-rounds (two full rounds), as shown here for a logical operator ($\bar{X}_0$ in Figure 7.6) of the Bolza Floquet code. Opposite sides of the tiling are identified. Crossing a bar of colour $c \in \{R,G,B\}$ means measuring the c-checks and then multiplying into each logical operator the c-checks that lie within its logical path. Note that the $\bar{X}_0$ logical operator (top left) is mapped to the $\bar{Z}_1$ logical operator (bottom right) every three sub-rounds and vice versa. This transition is an automorphism of $\mathrm{ISG}(R)$ which acts on each logical operator by multiplying by the inner logical operator that lies along the same logical path.

48 versus 96 subrounds. Six subrounds are two full rounds; therefore Figure 7.14's 16 rounds imply 48 subrounds. The public X/Z templates instead say 96 and contain REPEAT 15 of six ticks, with five prefix and two tail ticks (97 ticks total). A tick count alone does not reconcile the discrepancy. [1, pp. 162, 178; 3]

Figure 7 14 and numerical targets

Figure 7.14 including both panels and full caption | Printed page 178
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(a) **(b)**

Figure 7.14: Logical error rates vs. physical error rates for encoding 4 logical qubits using 4 copies of small planar honeycomb codes (blue) or a (semi-)hyperbolic floquet code encoding 4 logical qubits (red), for the EM3 noise model (check operators are measured directly without ancillas). In (a) we use a hyperbolic code derived from the Bolza surface that has distance 2 in this noise model, whereas in (b) we use an $l = 2$ semi-hyperbolic code derived from the Bolza surface that has distance 3. The legend gives the total number of qubits, with the dimension of the lattice given in parentheses for the planar honeycomb codes. All circuits use 16 rounds of measurements, and the y axis gives the average of the logical X and Z error rates. The black dotted line is the logical failure rate for the (semi-)hyperbolic code for a decoder which always predicts that no logical observable has been flipped. Note that the 4×6, 6×9 and 8×12 planar honeycomb codes have distances 2, 3 and 4 respectively for this EM3 noise model.

Figure derived values are not raw counts

At p = 0.001, the extracted block-error probabilities are approximately 0.1298 for four planar patches and 0.1339 for Bolza. The CSV contains 16 points per series, with coordinates and extraction bounds. Original trials, failures and separate X/Z values were not recovered.

Provenance: the arXiv v1 archive contains the original vector panel. All 32 colored marker centers align with the thesis panel after affine transformation; the maximum residual is 0.0000147 thesis PDF points. This is a direct version match, not an assumption that the versions coincide. [2]

Method: marker bounding-box centers and log-log axis inversion. A conservative +/-0.05 original-panel PDF-point tolerance gives about +/-0.0532% in p and +/-0.0414% in the logical error rate. These are extraction tolerances, not Monte Carlo confidence intervals. Exact anchors, transform and bounds are in the metadata and CSV.

Metric and aggregation

Figure 7.11 caption | Printed page 173 | PDF page 173 of 266 (1-based)

Figure 7.11: Threshold plots of two families of semi-hyperbolic Floquet codes, using an EM3 noise model and 64 rounds. (a) A family of semi-hyperbolic Floquet codes derived from the Bolza surface (see Table 7.2 for code parameters). (b) A family of semi-hyperbolic Floquet codes derived from a genus-33 closed hyperbolic surface (with code parameters given in Table 7.3). For both plots, the legend gives the fine-graining parameter l and number of physical qubits n and we observe a threshold of around 1.5%-2%, consistent with the threshold of the honeycomb code [97, 95, 156]. Note that here (and throughout this paper) the logical error rate is the probability that *any* of the encoded logical qubits fail.

Block failure means any encoded logical fails. It is distinct from the mean number of failed logicals divided by k. Figure 7.14 then averages the X- and Z-basis block-error rates. [1, pp. 173, 178]

Figure 7.12 caption | Printed page 175 | PDF page 175 of 266 (1-based)

Figure 7.12: Logical error rate vs. physical error rate for protecting 674 logical qubits using 674 copies of planar honeycomb codes (shades of blue) and a [[21504,674,12]] semi-hyperbolic Floquet code derived from a 8.8.8 tessellation. In (a) an EM3 noise model is used (direct pair measurements without the use of an ancilla) whereas in (b) an SD6 noise model (ancilla-assisted) is used and we also compare with standard surface code circuits (shades of green). For all codes, we simulate 192 time steps and give the average of the logical X and Z error rates on the y-axis. For the honeycomb and surface codes we plot $1-(1-p_{log})^{674}$ where p_{log} is the logical error rate for a single patch (one logical qubit). The legend gives the total number of physical qubits (including ancillas for the SD6 noise model).

Derived four-copy aggregation: for each basis b, $P_{4,b} = 1 - (1 - p_b)^4$, with the basis average $(P_{4,X} + P_{4,Z}) / 2$. The exponent 4 is this audit's application of the independent-copy method stated for 674 patches; it is not a verbatim equation in Figure 7.14. [1, pp. 175, 178]

Implementation context

The source uses Stim and PyMatching MWPM. In the EM3 templates, auxiliary qubit 16 implements pair measurement but is not a seventeenth physical hardware qubit. The repository warns that supplied Bolza logical representatives differ from the figure and are not symplectic; logical-space equivalence is therefore not automatic. [1, p. 167; 3]

Sources and artifact provenance

[1] Higgott thesis PDF, UCL Discovery. 266 pages. [2] arXiv 2308.03750v1 record and source archive. Vector member: figures/bolza_l1_vs_planar_honeycomb_em3.pdf.

[3] oscarhiggott/hyperbolic-floquet-data, pinned commit 457975f0d6c67b3a505aacf44589d36779f5dcc1. [4] Strilanc/honeycomb-boundaries, pinned commit cc33baac44c7831bd643db81d0053f8ec6eae9d8.

The checked repository trees and arXiv source archive did not yield Figure 7.14 raw shot counts or numerical arrays. The publisher record (PRX Quantum 5, 040327) points to the same repositories. This search result does not prove that raw data do not exist.

SHA256 integrity index

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